that increased capacity of the skull is one evidence of degeneration, development in size being in inverse ratio to development in structure. In general paralysis the skulls examined (of males) were a little above the healthy average. This Dr. Amadei ascribes to the patients having been of a higher class in life, a source of fallacy which is not to be forgotten.

Dr. Buccola has been measuring the *Rapidity of Dilatation of the Iris* after puncturing one hand or arm, employing an apparatus to register the moment at which the pupil is found to dilate. He thus examined 15 cases of general paralysis (13 males and 2 females); in seven the pupils were found not to act at all, while in the remainder the movement was considerably delayed. This appears not to be the case in ordinary insanity, where the period of re-action was found to be the same as in health.

The longest and most elaborate paper in the *Rivista* is one by Prof. Golgi, on *The Minute Anatomy of the Nerve-Centres.* His object is to discover an anatomical basis for some of the data of physiology, selecting particularly the following problems:—1. The connection between the nerve-fibres and the ganglionic cells. 2. The relation between the different forms of ganglionic cells and the functions of the parts where they are found. 3. The arrangement and relations of the elements in the several parts of the nerve-centres. 4. The course of the nerve-fibres and their relation to the groups of ganglionic cells. Space only allows of my extracting a few of the most notable points dealt with in this remarkable essay. He defines nerve-cells to be those only which are provided with a single prolongation, continuous with the fibres. The "protoplasmatic prolongations" of Deiters are not connected with the nerve-fibres, as may be most clearly seen in the fascia dentata. Golgi suggests that they are probably the means by which nutrition is kept up in the cells.

The nerve-prolongation sends off at tolerably regular intervals lateral filaments, which in turn divide and subdivide into an extremely complicated network through all the grey matter. Similar filaments seem to be given off from the ordinary nerve-fibres. There seem, therefore, to be two different ways in which the fibres and cells are connected; either directly by a prolongation from a nerve-cell becoming the cylinder-axis of a fibre, or indirectly by the filaments given off from the prolongations and fibres, which are closely interlaced in the grey matter. Taking the case of the spinal cord, the direct connection seems to prevail especially in the anterior cornua; the indirect connection in the posterior cornua: the former being therefore prevalent in the motor centres, the latter in the sensory.

For the most part each nerve-fibre is connected with separate groups of ganglionic cells; and each ganglionic cell in turn, with nerve-fibres, going in different directions.

His examination of the cortical grey matter leads him to differ from Meynert, and to divide it into the following layers, which gradually shade into each other:—1. An outer layer, composed principally of

rather small pyramidal cells. 2. A middle layer of larger cells, also pyramidal. 3. The inner layer, the cells in which are mostly fusiform, but also polygonal, globose, or irregular in the anterior convolutions, but containing many small cells in the deepest part of this layer in the occipital convolutions. This is the only histological distinction noted by our author in different parts of the convolutions.

We have received the *Archives Italiennes de Biologie* for 1882, a periodical published at Turin, and intended to supply abstracts in French of all the more notable biological papers published in Italy. The only paper bearing on our subject, which has not been already noticed in these Retrospects, is one by Dr. Marcacci, on the *Cortical Motor Centres*. He is led by a careful analysis of 27 cases to conclude that there is no pathological evidence of the existence of localized motor centres in man.

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2. *French Retrospect.*

By Dr. T. W. McDowall.

Les Annales Médico-Psychologiques. Sept., 1880, to Jan., 1881.

The Treatment of Delirium Tremens and of Acute Delirium by Cold Baths and Bromide of Potassium. By Dr. Rosseau.

This short paper essentially consists in the record of two cases of acute delirium.

In the first, the patient was admitted in a state of extreme excitement, the face red, the eyes bright, the temperature increased, the pulse very rapid, full, regular. He had hallucinations, and the delusions usually found in such cases. He was immediately ordered 10 grammes of bromide of potassium, but the excitement and delirium continued during the day and the following night.

Next day the bromide was continued; but in addition, the patient was kept in a cool bath for seven hours, and from time to time cold compresses were applied to the forehead. During this process the excitement and violence disappeared, though his sanity was not completely restored. He had an excellent night, and slept till morning.

On 21st July he was reported to be free from excitement, and quite rational. The baths were discontinued, but the bromide was continued until the 23rd as a precautionary measure. In five days he consumed 50 grammes of that drug. He was discharged on the 29th, quite well.

The second case was admitted on the 18th November, 1879. He suffered from acute delirium, such as is so frequently seen in the beginning of general paralysis. He was 32 years of age, had always been sober in his habits, and there was no history of insanity in his family. He had experienced reverses in business, had led a sedentary life, and was of an exceedingly sanguine temperament.